

Ryan Sullivan

Senior Full Stack & Mobile Software Engineer

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SUMMARY

Senior Full-Stack Engineer with 10+ years of expertise in scalable microservices, composable architectures, and cross-platform mobile development across AWS, Azure, and GCP. Strong background in TypeScript, React, Next.js, Node.js, and React Native, delivering high-performance web and mobile applications with real-time capabilities and optimized Core Web Vitals. Experienced in event-driven systems (Kafka, RabbitMQ), advanced search architectures (Elasticsearch, Algolia), distributed caching (Redis), and Kubernetes-based deployments. Skilled in secure authentication, Stripe subscription systems, and AI integrations (OpenAI, GitHub Copilot, Claude) for semantic enrichment and automation. Proven track record of leading large-scale platform migrations and delivering production-grade systems in Agile environments.

EDUCATION

Bachelor’s Degree in Computer ScienceAug 2012 – May 2016

University of Florida | Gainesville, FL

EXPERIENCE

Senior Full Stack & Mobile Software EngineerApr 2023 – Mar 2026

Orangesoft | San Francisco, CA

- Led migration of a 100K+ page medical platform to a composable architecture using **TypeScript**, **Next.js**, **React**, **Node.js**, Sanity, Algolia, Elasticsearch, **Redis**, Docker, Kafka, **RabbitMQ**, Vercel, and **AWS (S3, CloudFront, IAM, Route 53, EKS)**, integrating Stripe for subscription management.
- Architected scalable, multi-region content models in **Sanity Studio** v3 (**GROQ**, custom schemas, RBAC, **i18n**, **webhooks**) enabling structured editorial workflows and automated publishing pipelines.
- Designed **REST** and **gRPC** backend services with **JWT** authentication and **RBAC**, leveraging **Kafka**, RabbitMQ, and **WebSocket** for event-driven processing and real-time communication., backed by optimized **PostgreSQL** schemas and indexing strategies.
- Implemented advanced search architecture with **Elasticsearch** and **Algolia**, building unified indexing pipelines, relevance tuning, and webhook-driven reindexing across 40K+ clinical articles and multimedia assets.
- Improved system performance through **Redis caching**, distributed session management, and optimized database queries under high traffic conditions.
- Containerized **microservices** using **Docker** and orchestrated with **Kubernetes (EKS)**, enabling horizontal scaling and zero-downtime deployments.
- Implemented hybrid **SSG**, **SSR**, and **ISR** using Next.js App Router, API Routes, and Edge Functions, deployed via **Vercel** and **AWS**, achieving green Core Web Vitals (LCP, INP, CLS).
- Optimized frontend performance using dynamic imports, React.memo, Tailwind CSS, WebSocket-driven live updates, and CloudFront edge caching.
- Engineered secure subscription flows with **Stripe API**, **Apple Pay**, and **Google Pay**, enforcing server-side validation, **AWS IAM** policies, and compliance-grade audit logging.
- Built cross-platform mobile applications with **React Native**, **Expo**, and Redux Toolkit, supporting CPD tracking, real-time updates via WebSocket, and offline-first synchronization.
- Integrated **GenAI** capabilities using **OpenAI GPT-4o**, **GitHub Copilot** and **Anthropic Claude APIs** for clinical summarization, semantic enrichment, structured data extraction, and AI-assisted editorial workflows with structured output validation and guardrails.
- Implemented observability and monitoring with **Grafana**, centralized logging, and CI/CD automation via **Expo CLI**, **EAS Build**, and **GitHub Actions**, delivering 21 epics in a Scrum-based Agile environment.

Senior Mobile & Full Stack Software Engineer

Jul 2020 – Mar 2023

Tepia | Costa Mesa, CA

- Delivered a cross-platform mobile and web MVP using TypeScript, **React Native**, **Expo**, **React.js**, **Nest.js**, **Firebase**, **Sanity CMS**, and **Azure Cloud Services**.
- Built iOS and Android applications with structured state management and reusable UI components using React Native, Expo, Redux Toolkit, and Styled Components.
- Developed responsive frontend with React.js, TypeScript, React Router, and **Axios**, deployed via Azure App Service and **Azure Static Web Apps**.
- Engineered backend APIs and serverless business logic using **Nest.js**, **Firebase Cloud Functions**, and Azure Functions for scalable, modular, and maintainable microservices architecture.
- Implemented real-time data sync and secure storage using **Firebase Authentication**, **Firestore**, **Azure Blob Storage**, and security rules.
- Integrated dynamic content delivery with Sanity CMS, **GROQ**, **YouTube API**, optimized via **Azure CDN**.
- Built push notification workflows using Expo Notifications, **Firebase Cloud Messaging**, **Azure Notification Hubs**, and Cloud Scheduler.
- Implemented analytics via Firebase Analytics, Azure Application Insights, and custom event tracking.
- Automated lifecycle workflows using **Mailchimp API**, HTTPS Functions, and **Azure Logic Apps**.
- Configured CI/CD with **Expo EAS Build**, **GitHub Actions**, and **Azure DevOps** Pipelines.

Mobile Software Engineer

Oct 2018 – Jun 2020

Mercury Development | Aventura, FL

- Architected scalable cross-platform mobile applications using **React Native**, **Expo**, and **TypeScript**, implementing modular architecture with **React Navigation** and **Context API** for maintainability and rapid iteration.
- Engineered secure authentication flows including passwordless Magic Link, **JWT**, and **OAuth**, with secure storage via Expo SecureStore and role-based access control across **GraphQL** (Apollo Client) services.
- Developed QR-based voucher scanning with Expo Camera, deep linking, background tasks, and real-time validation, ensuring low-latency and secure transaction processing.
- Optimized performance through efficient data-fetching strategies, intelligent Apollo caching, optimistic UI updates, memoization, lazy loading, and bundle optimization.
- Built reusable component libraries in **Storybook** following atomic design from **Figma**, improving UI consistency and development velocity.
- Established CI/CD pipelines with GitHub Actions and Expo EAS Build; managed releases via **App Store Connect** and **Google Play Console**, enforcing code quality with **ESLint**, **Prettier**, **Husky**, **Jest**, and **React Native Testing Library**.

Full Stack Software Engineer

Jun 2016 – Sep 2018

Culture Foundry | Plano, TX

- Designed and deployed a microservice-based onboarding application on **Google Cloud Platform** (GCP) using **Cloud Run** and **Cloud Build**, enabling scalable, containerized **CI/CD**-driven releases.
- Built the frontend with **Next.js**, **TypeScript**, **Tailwind CSS**, and **Storybook**, developing reusable component libraries and enforcing modular, maintainable architecture.
- Implemented secure **API integrations** without modifying the existing **Amplience API**, ensuring scalable and fault-tolerant cloud communication.
- Led structured code reviews, enforcing clean architecture, **SOLID principles**, and TypeScript best practices across the JavaScript codebase.
- Established automated testing pipelines with **Jest**, **Mocha**, and React Testing Library, integrating coverage checks, linting (**ESLint**, **Prettier**, **Husky**), and CI quality gates.
- Delivered iterative releases within **Scrum** using GCP-backed automated build, test, and deployment workflows.

SKILLS

Programming Languages:

TypeScript, JavaScript (ES6+), SQL, GraphQL, GROQ

Backend & APIs:

Node.js, Nest.js, REST, gRPC, GraphQL (Apollo), JWT Authentication, RBAC, Webhooks, Serverless Architecture

Frontend & Web:

React.js, Next.js (App Router, SSR, SSG, ISR), Tailwind CSS, Styled Components, React Router, Axios, Storybook, WebSocket, Core Web Vitals Optimization

Mobile Development:

React Native, Expo, Redux Toolkit, Context API, React Navigation, Expo EAS Build, Push Notifications (FCM, Expo), Deep Linking, Background Tasks, Offline-First Architecture

Databases & Caching:

PostgreSQL, MySQL, Oracle, Firestore, Redis, Elasticsearch, Algolia

Cloud & DevOps:

AWS (S3, CloudFront, IAM, Route 53, EKS), Azure (App Service, Static Web Apps, Blob Storage, Functions, Notification Hubs), Google Cloud Platform (Cloud Run, Cloud Build), Docker, Kubernetes, CI/CD (GitHub Actions, Azure DevOps), Vercel

Architecture & Systems:

Microservices, Event-Driven Architecture, Kafka, RabbitMQ, Distributed Systems, Multi-Region Deployment, Edge Computing, Horizontal Scaling

CMS & Content Platforms:

Sanity Studio v3 (Custom Schemas, GROQ, RBAC, i18n, Webhooks), Amplience

Payments & Integrations:

Stripe API, Apple Pay, Google Pay, Mailchimp API, YouTube API

AI & Automation:

OpenAI GPT-4o API, GitHub Copilot, Anthropic Claude API, AI-Assisted Workflows, Structured Output Validation, Prompt Engineering

Monitoring & Analytics:

Grafana, Firebase Analytics, Azure Application Insights, Centralized Logging

Testing & Code Quality:

Jest, Mocha, React/React Native Testing Library, ESLint, Prettier, Husky

Methodologies:

Agile / Scrum, CI/CD Automation, Clean Architecture, SOLID Principles, Code Reviews

PROJECTS

Healthcare & Medical Education Platform

<https://GPnotebook.com>

Modernized and scaled GPnotebook, a 100K+ page clinical reference platform, by migrating from a legacy CMS to a composable architecture built with Next.js, React, TypeScript, Node.js, Sanity, Algolia, Stripe, and Vercel, enabling high-performance delivery, multi-region content management, CPD compliance tracking, subscription monetization, and unified web and mobile experiences for medical professionals across the UK, US, Australia, and Ireland.

InsurTech Full MVP Platform

<https://veygo.com>

Led development of a cross-platform mobile and web MVP using React Native, Expo, React.js, TypeScript, Node.js, Firebase, and Sanity CMS to validate a new InsurTech learning product. The platform enabled progress tracking, personalized content delivery, push notifications, analytics-driven user behavior insights, and secure customer data handling to support rapid idea validation and risk mitigation